

Student Performance Data Cleaning & Visualization

Data Analytics Project

Objective: Clean a raw student-performance dataset, analyze academic outcomes, identify patterns, and communicate insights through visualizations.

1. Dataset Description

The dataset contains 1,000 student records and 8 variables covering gender, race/ethnicity, parental education, lunch type, test preparation, and scores in math, reading, and writing. The dataset is suitable for exploratory data analysis because it combines categorical and numerical variables.

2. Data Cleaning

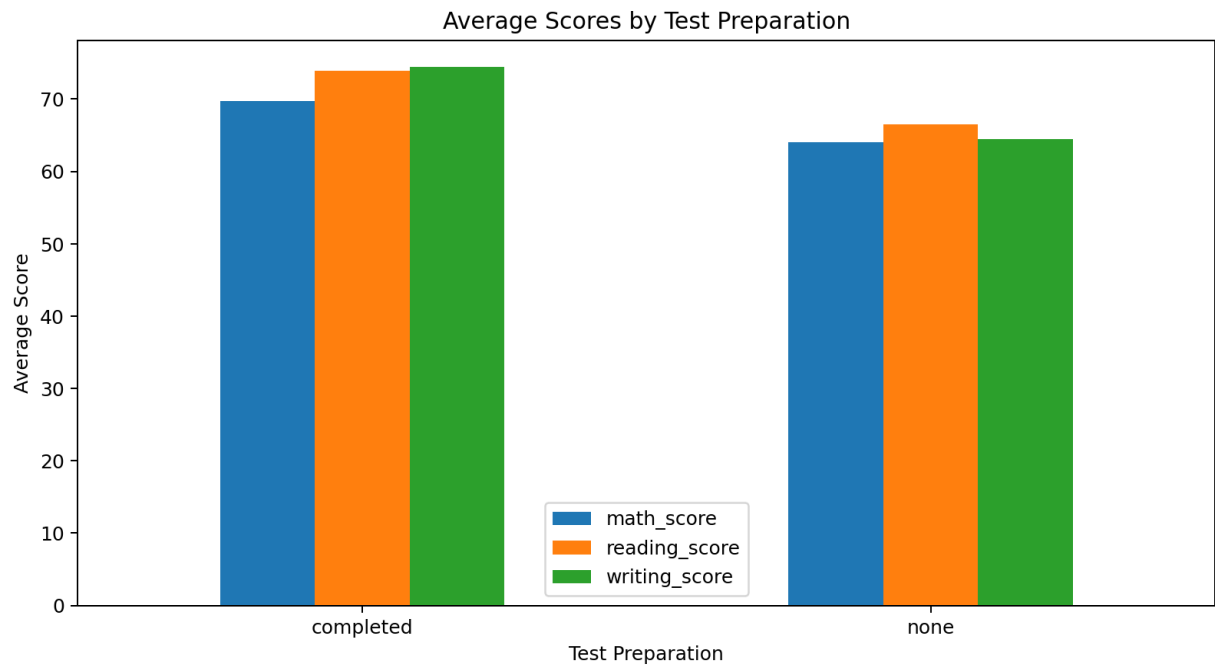
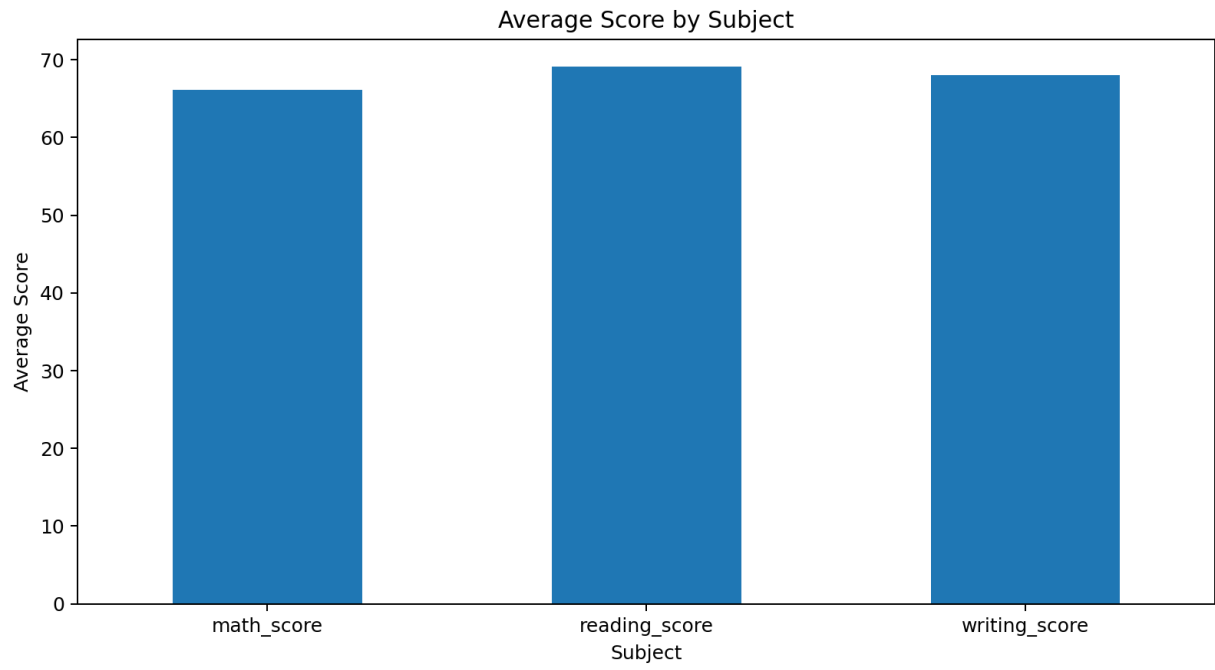
Check	Result	Action
Rows before cleaning	1,000	Reviewed
Missing values	0	No missing values found
Duplicate rows	0	No exact duplicates found
Invalid scores	0	All scores are within 0–100

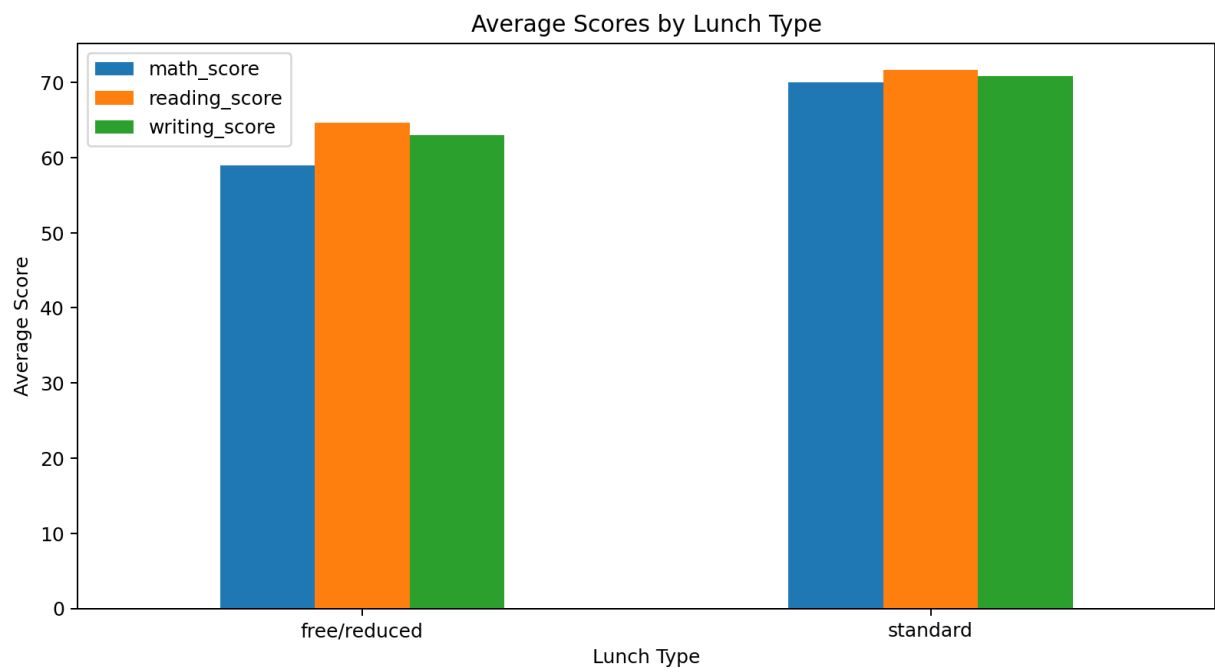
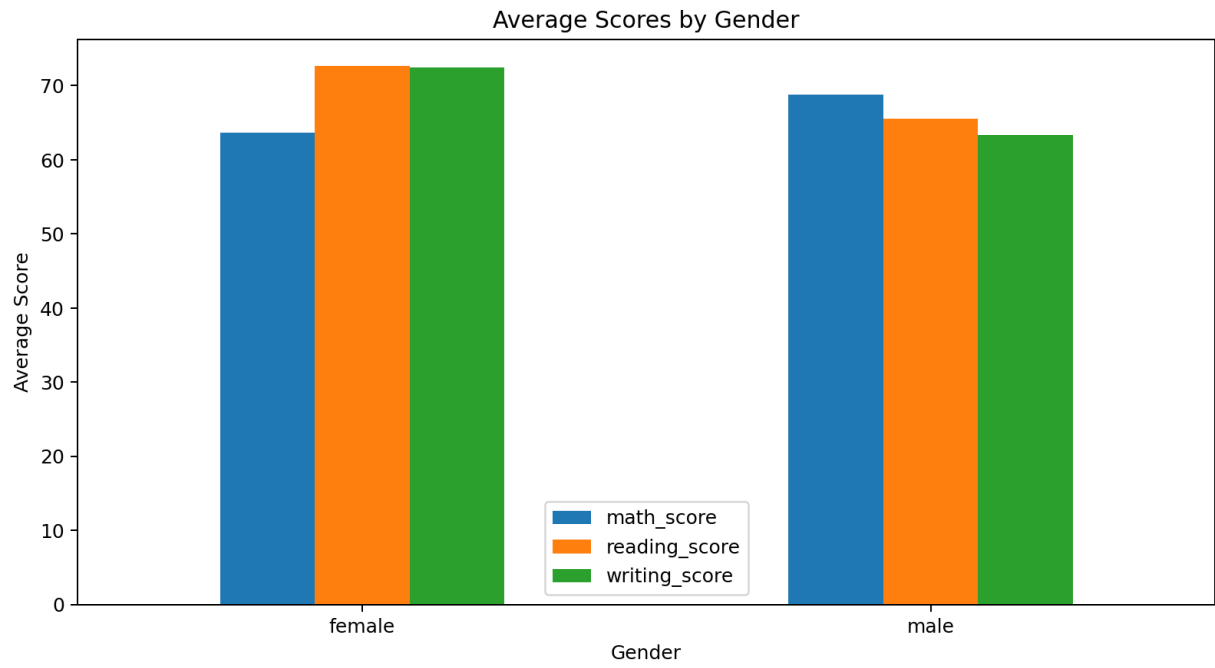
Potential outliers were detected with the IQR method. Because the score values are valid observations on a 0–100 exam scale, they were retained rather than removed as errors.

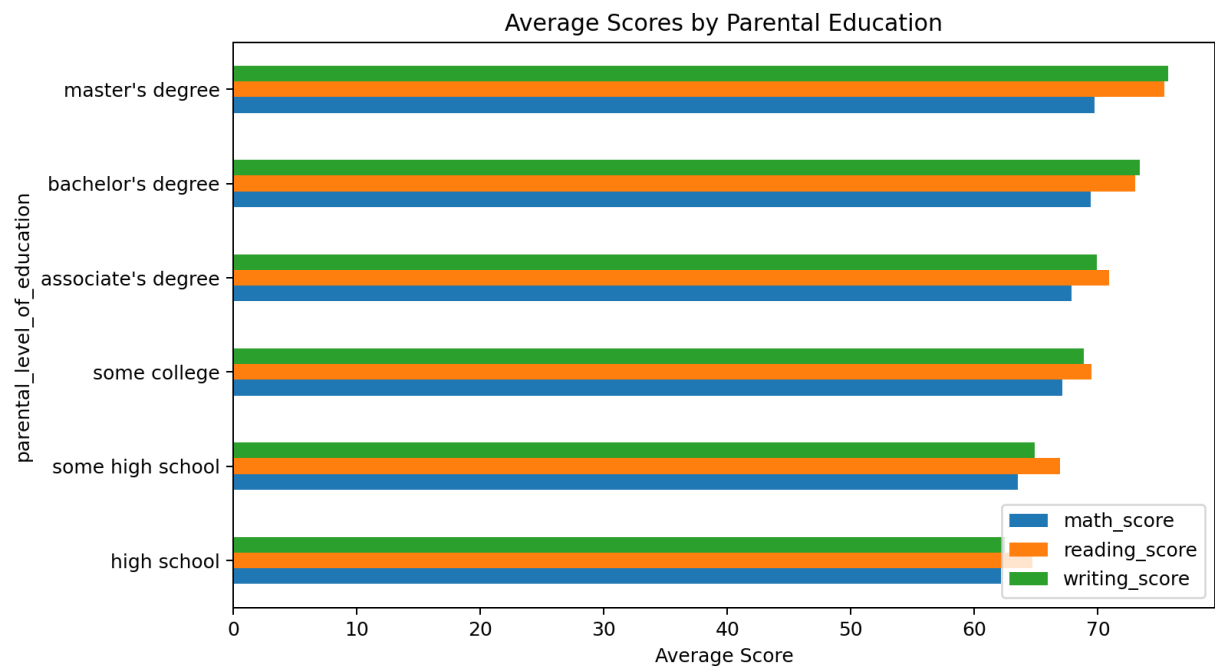
3. Descriptive Analysis

Subject	Mean	Median	Minimum	Maximum
Math Score	66.09	66.00	0	100
Reading Score	69.17	70.00	17	100
Writing Score	68.05	69.00	10	100

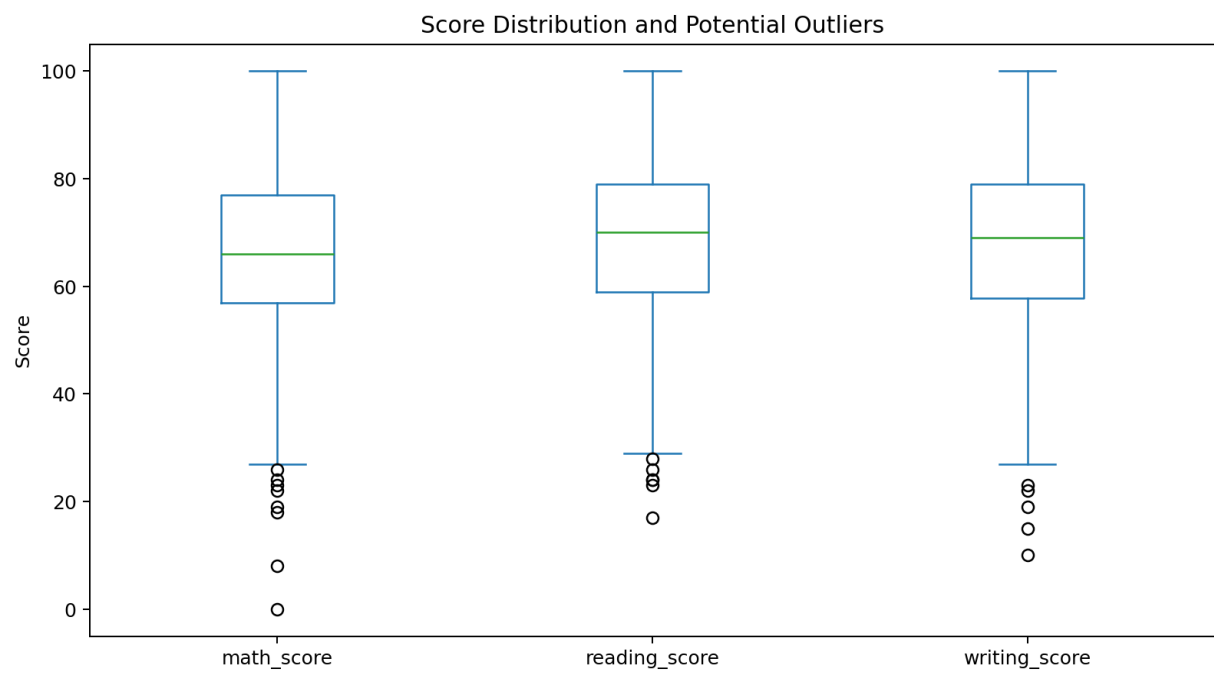
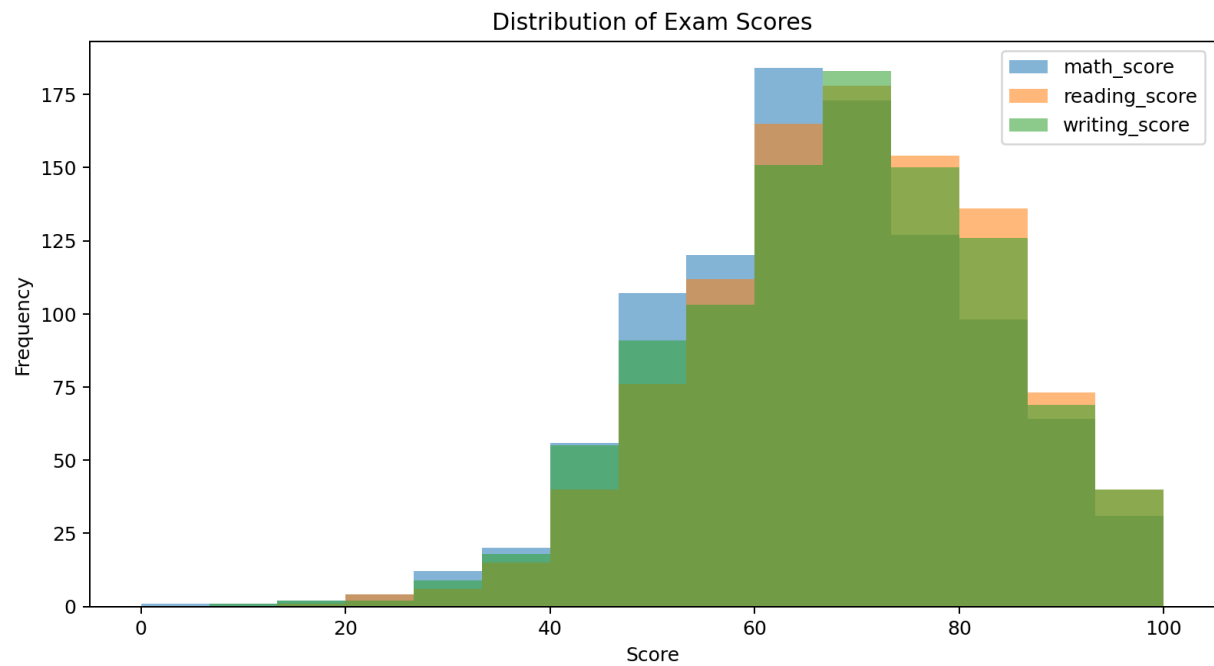
4. Visual Analysis

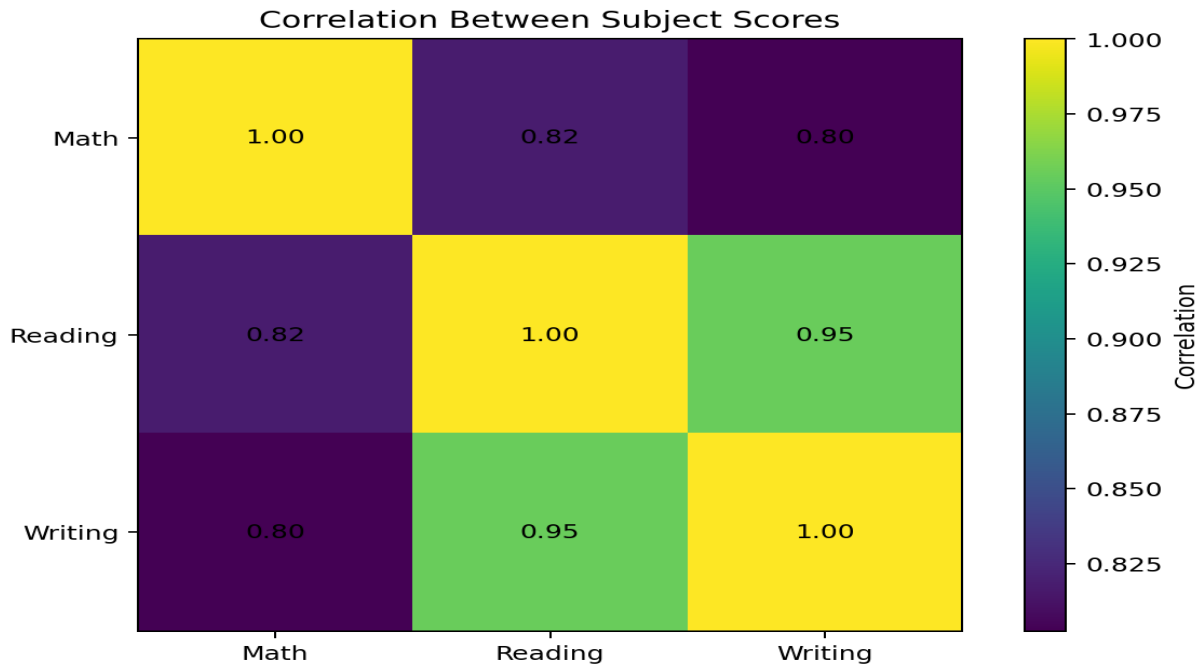






5. Additional Visualizations





6. Key Findings

- Reading has the highest overall average score (69.17), followed by writing (68.05) and mathematics (66.09).
- Students who completed the test preparation course have a higher average overall score (72.67) than students who did not (65.04).
- Students receiving standard lunch have a higher average overall score (70.84) than students receiving free/reduced lunch (62.20).
- Reading and writing scores show a strong positive relationship, indicating that students who perform well in one language-related subject tend to perform well in the other.
- Potential outliers identified by the IQR method were reviewed and retained because they represent valid exam scores rather than impossible values.

7. Conclusion

The project demonstrates a complete data-analysis workflow: loading raw data, checking data quality, cleaning records, creating derived measures, analyzing group differences, and presenting results visually. The analysis suggests that test preparation and lunch category are associated with differences in average performance, while the three subject scores are positively related. The visualizations make these patterns easier to understand and communicate.

8. Tools Used

Python, Pandas, Matplotlib, Seaborn, Jupyter/Google Colab, and Excel.